

Question 1 (12 marks) **Marks**

- a) Solve $\frac{3}{x-1} < 4$. **3**
- b) Find $\lim_{x \rightarrow 0} \frac{\sin x}{5x}$. **1**
- c) Given $\sin \theta + \cos \theta = A$ and $\sin \theta \cos \theta = B$. Express $\sin^3 \theta + \cos^3 \theta$ in terms of A and B . **2**
- d) Find $\int \frac{\cos \theta}{5 - \sin \theta} d\theta$ **2**
- e) The two parabolas $y = x^2$ and $y = (x-2)^2$ intersect at the point $(1,1)$. Find the angle between the tangents to the curves at $x = 1$. Give your answer to the nearest degree. **4**

Question 2 (12 marks) Start this question in a new booklet.

- a) Use the substitution $u = 9 - x^2$ to find $\int_0^1 6x\sqrt{9-x^2} dx$. **4**
- b) Given $\frac{d}{dx}(x \sin x) = \sin x + x \cos x$, find $\int x \cos x dx$. **2**
- c) (i) State the domain and range of the function $y = 3 \sin^{-1}\left(\frac{x}{2}\right)$ **2**
 (ii) Draw a sketch of the function, carefully labeling the extremities of both the range and domain. **1**
- d) The letters of the word *CALCULUS* are arranged in a row. **3**
 (i) How many different arrangements are there?
 (ii) If one of these arrangements is selected at random, find the probability that it begins with "U" and ends with a "U".

Question 3 (12 marks) Start this question in a new booklet. **Marks**

- a) Consider the parabola $x^2 = 4ay$ where $a > 0$, and suppose the tangents at $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ intersect at the point T .
- (i) Given the equation of the tangent from P is $y = px - ap^2$, **2**
show that the point T has coordinates, $(a(p + q), apq)$
- (ii) Given S is the focus, show that $SP = a(p^2 + 1)$. **2**
- (iii) If P and Q move along the parabola such that $SP + SQ = 4a$, **2**
find the equation of the locus of point T .
- b) The function $f(x) = x - e^{-x}$ has one root between 0 and 1.
- (i) Show that the root lies between 0.4 and 0.6. **1**
- (ii) Use one application of Newton's Method with an **2**
initial approximation of $x = 0.5$ to find a second approximation.
- c) A polynomial $f(x)$ leaves a remainder of -10 when divided by **3**
 $(x - 1)$ and 20 when divided by $(x + 2)$. If $f(x) = x^n + ax - 6$, show that $x + 1$ is a factor.

Question 4 (12 marks) Start this question in a new booklet.

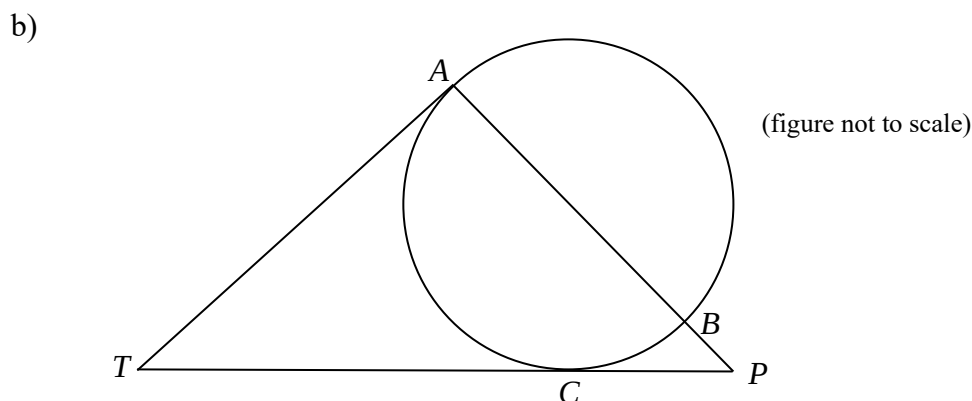
- a) The speed v cm/s of a particle with simple harmonic motion in a straight line is given by $v^2 = 6 + 4x - 2x^2$, where x cm is the magnitude of the displacement from a fixed point O .
- (i) Show that $\ddot{x} = -2(x - 1)$. **2**
- (ii) Find the centre of the motion. **1**
- (iii) Find the period of the motion. **1**
- (iv) Find the amplitude of the motion. **2**

Question 4 continued**Marks**

- b) Find, as a rational number, the coefficient of x in the expansion of $\left(x^2 + \frac{1}{2x}\right)^8$. **3**
- c) Colour blindness affects 5% of all men. What is the probability that any random sample of 20 men should contain:
- (i) no colour blind men. **1**
- (ii) exactly 4 colour blind men. **2**

Question 5 (12 marks) Start this question in a new booklet.

- a) (i) Show that $\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cos^2 x \, dx = \frac{\pi}{8} - \frac{1}{4}$. **2**
- (ii) Hence or otherwise find the volume generated when $y = 1 - \cos x$ is rotated about the x axis between $x = \frac{\pi}{4}$ and $x = \frac{\pi}{2}$. **2**



AB is a diameter of a circle ABC . The tangents at A and C meet at T . The lines TC and AB are produced to meet at P .

Copy the diagram into your examination booklet. Join AC and CB .

- (i) Prove that $\angle CAT = 90^\circ - \angle BCP$. **2**
- (ii) Hence, or otherwise prove that $\angle ATC = 2\angle BCP$. **2**

Question 5 continued**Marks**

- c) At time t the temperature T° of a body in a room of constant temperature 20° is decreasing according to the equation $\frac{dT}{dt} = -k(T - 20)$ for some constant $k > 0$.
- (i) Verify that $T = 20 + Ae^{-kt}$, where A is a constant, is a solution of the equation. **1**
- (ii) The initial temperature of the body is 90° and it falls to 70° after 10 minutes. Find the temperature of the body after a further 5 minutes. (Answer to the nearest degree.) **3**

Question 6 (12 marks) Start this question in a new booklet.

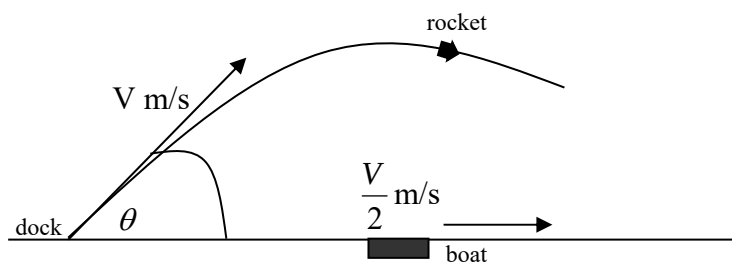
- a) Consider the function $f(x) = \frac{e^x}{1 + e^x}$.
- (i) Show that $f(x)$ is increasing for all values of x . **2**
- (ii) Explain why $f(x)$ has an inverse function. **1**
- (iii) Find the inverse function $y = f^{-1}(x)$. **2**
- b) A yacht is sailing due north off the west coast of a small island with an extinct volcanic peak of height 1200m.
At 0900 hours the bearing of the peak is 010° and the angle of elevation 5° . At 1030 hours the bearing of the peak is 120° and the angle of elevation 20° .
- (i) Draw a diagram to illustrate the information above. **1**
- (ii) Calculate their speed in km/h. **3**
- c) (i) Express $\sin x + \sqrt{3} \cos x$ in the form $R \sin(x + \alpha)$. **1**
- (ii) Hence solve $\sin x + \sqrt{3} \cos x = 1$ for $0 \leq x \leq 2\pi$ **2**

Question 7 (12 marks) Start this question in a new booklet.**Marks**

- a) A boat leaves a dock at a speed of $\frac{V}{2}$ m/s and travels in a straight line. A rocket is fired simultaneously from the same starting point and in the same direction, with initial speed V m/s and angle of elevation θ , where $0^\circ < \theta < 90^\circ$. Also assume the height of the dock and the boat is zero metres.

Given the rocket's trajectory is defined by the equations:

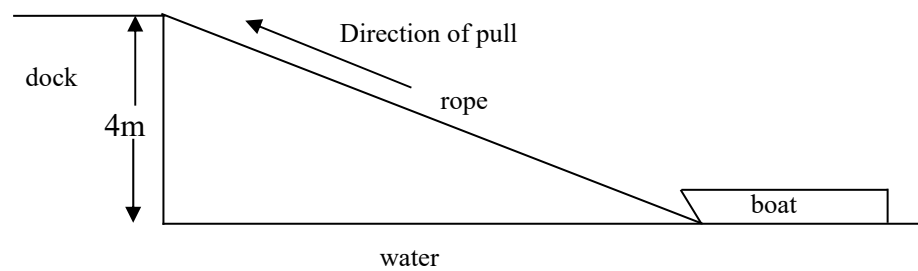
$$x = Vt \cos \theta \text{ and } y = -\frac{1}{2}gt^2 + Vt \sin \theta$$



- (i) show that the time of flight of the rocket is 2
 $\frac{2V \sin \theta}{g}$ seconds.
- (ii) show that the range of the rocket is 1
 $\frac{2V^2 \sin \theta \cos \theta}{g}$ metres.
- (iii) determine the angle of elevation of the rocket needed 2
 for the rocket to hit the boat
- b) Use mathematical induction to prove that, for every positive 3
 integer n , $9^n - 3$ is divisible by 6.

Question 7 continued**Marks**

- c) A boat is pulled towards a dock by a rope through a ring on the dock which is 4 metres above the boat. If the rope is pulled in at 2 m/s, how fast is the boat travelling when the rope is 10 metres long? **4**

**End of paper.**

① (a) $\frac{3}{x-1} \times (x-1)^2 < 4(x-1)^2$

$$3(x-1) < 4(x-1)^2$$

$$4(x-1)^2 - 3(x-1) > 0$$

$$(x-1)(4x-7) > 0$$

$$x < 1 \text{ or } x > \frac{7}{4}$$

(b) $\frac{1}{5}$

(c) $\sin^3 \theta + \cos^3 \theta$

$$= (\sin \theta + \cos \theta)(\sin^2 \theta - \sin \theta \cos \theta + \cos^2 \theta)$$

$$= A(1-B)$$

or

$$(\sin \theta + \cos \theta)^3 = \sin^3 \theta + 3\sin^2 \theta \cos \theta + 3\sin \theta \cos^2 \theta + \cos^3 \theta$$

$$\therefore \sin^3 \theta + \cos^3 \theta = (\sin \theta + \cos \theta)^3 + 3\sin \theta \cos \theta (\sin \theta + \cos \theta)$$
$$= A^3 - 3BA$$

(d) $-\ln(5 - \sin \theta) + C$

(e) Let θ = acute angle between lines

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$m_1 = 2x, \text{ at } x=1$$
$$= 2$$

$$m_2 = 2(x-2) \text{ at } x=1$$
$$= -2$$

$$\tan \theta = \left| \frac{2 - (-2)}{1 + 2 \cdot (-2)} \right|$$

$$= \left| -\frac{4}{3} \right|$$

$$\theta = 53^\circ \text{ (nearest degree)}$$

$$2(a) \int_0^1 6x\sqrt{9-x^2} dx$$

$$= -3 \int_0^1 \sqrt{9-x^2} (-2x dx)$$

$$= -3 \int_9^8 u^{1/2} du$$

$$= -3 \left[\frac{2}{3} u^{3/2} \right]_9^8$$

$$= -3 \left(\cancel{\frac{2}{3}} \frac{2}{3} \sqrt{512} - 18 \right)$$

$$= -2\sqrt{512} + 54$$

$$= -32\sqrt{2} + 54$$

$$u = 9 - x^2$$

$$\therefore du = -2x dx$$

$$\text{if } x=1, u=8$$

$$x=0, u=9$$

$$8.745166004$$

$$(b) \frac{d}{dx} (x \sin x) = \sin x + x \cos x$$

$$\therefore x \cos x = \frac{d}{dx} (x \sin x) - \sin x$$

$$\therefore \int x \cos x dx = \int \left[\frac{d}{dx} (x \sin x) - \sin x \right] dx$$

$$= x \sin x - \int \sin x dx$$

$$= x \sin x + \cos x + C$$

$$(c) (i) y = 3 \sin^{-1} \left(\frac{x}{2} \right)$$

$$\therefore \frac{1}{3} y = \sin^{-1} \left(\frac{x}{2} \right)$$

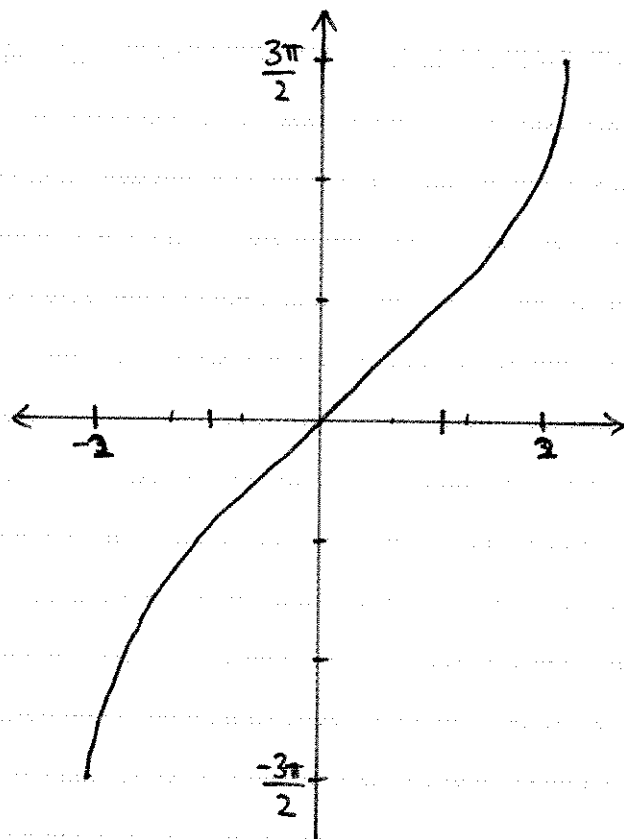
$$\therefore \sin \frac{1}{3} y = \frac{x}{2}$$

$$\therefore 2 \sin \frac{1}{3} y = x$$

$$\therefore -2 \leq x \leq 2$$

$$-\frac{3\pi}{2} \leq y \leq \frac{3\pi}{2}$$

(ii)



$$(d)(i) \text{ Arrangements} = \frac{8!}{2!2!2!} = 5040$$

$$(ii) \text{ Probability} = \frac{6!}{2!2!} = \frac{180}{5040}$$

$$= \frac{1}{28}$$

$$= \frac{1}{28}$$

for simpl. from factorial form

$$\text{OR Prob} = (\text{Prob } 1^{\text{st}} \text{ letter is a U}) \times (\text{Prob. last letter is a U})$$

$$= \frac{2}{8} \times \frac{1}{7}$$

$$= \frac{1}{28}$$

$$(3) (a) (i) y = px - ap^2 \text{ ——— } (1)$$

$$y = qx - aq^2 \text{ ——— } (2)$$

Solve (1), (2) simultaneously

$$x = a(p+q), y = apq$$

$$(ii) S(0, a), P(2ap, ap^2)$$

$$SP = \sqrt{(2ap-0)^2 + (ap^2-a)^2}$$

$$= \sqrt{4a^2p^2 + a^2p^4 - 2a^2p^2 + a^2}$$

$$= \sqrt{a^2(p^2+1)^2}$$

$$= a(p^2+1)$$

$$(iii) SP = a(p^2+1), SQ = a(q^2+1)$$

$$a(p^2+1) + a(q^2+1) = 4a$$

$$p^2 + q^2 = 2 \text{ ——— } (3)$$

$$x = a(p+q), y = apq$$

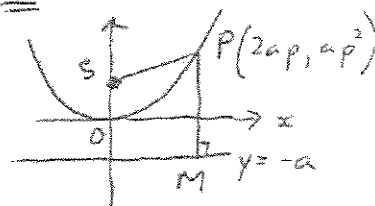
$$x^2 = a^2(p^2 + q^2 + 2pq)$$

$$= a^2\left(2 + \frac{2apq}{a}\right) \text{ from (3)}$$

$$= a^2\left(2 + \frac{2y}{a}\right)$$

$$= 2a(a + y)$$

or



$$SP = PM \text{ (by definition)}$$

$$= ap^2 - (-a)$$

$$= ap^2 + a = a(p^2+1)$$

$$(4) (i) f(0.4) < 0, f(0.6) > 0$$

or

$$f(0.4) \cdot f(0.6) < 0$$

$\therefore$ root exists

between 0.4 & 0.6

$$\begin{aligned}
 \text{b (ii)} \quad x_2 &= x_1 - \frac{f(x_1)}{f'(x_1)} \\
 &= x_1 - \frac{x_1 - e^{-x_1}}{1 + e^{-x_1}}
 \end{aligned}$$

$$x_1 = 0.5$$

$$\therefore x_2 \approx 0.57$$

$$(c) \quad f(1) = -10, \quad f(2) = 20$$

$$\therefore 1^n + a - 6 = -10$$

$$a = -5$$

$$(-2)^n + 10 - 6 = 20$$

$$(-2)^n = 16$$

$$n = 4$$

$$\therefore f(x) = x^4 - 5x - 6$$

$$f(-1) = 1 + 5 - 6$$

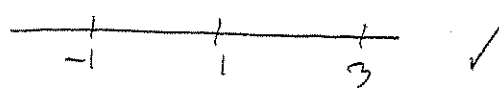
$$= 0$$

$\therefore x+1$ is a factor.

Question 4

$$\begin{aligned}
 \text{a) i) } \ddot{x} &= \frac{d}{dt} \left(\frac{1}{2} v^2 \right) = \frac{d}{dt} \left(\frac{1}{2} (6 + 4x - 2x^2) \right) \checkmark \\
 &= \frac{d}{dt} (3 + 2x - x^2) \checkmark \\
 &= 2 - 2x \\
 &= -2(x-1)
 \end{aligned}$$

ii) clearly centre is $x=1$.
 $V=0$ at end points
 $\therefore 0 = 6 + 4x - 2x^2$
 $0 = x^2 - 2x - 3$
 $0 = (x-3)(x+1)$
 $x = -1$ or 3



$\therefore$ centre of motion is $x=1$

iii) $n^2 = 2$
 $n = \sqrt{2}$
 Period $= \frac{2\pi}{n}$
 $= \frac{2\pi}{\sqrt{2}}$
 $= \sqrt{2} \pi \checkmark$

iv) From part ii
 amplitude $= 2 \checkmark$

Question 4

b) $(x^2 + \frac{1}{2x})^8 = \sum_{r=0}^8 (x^2)^r \left(\frac{1}{2x}\right)^{8-r} {}^8C_r \checkmark$
 $(x^2)^r \left(\frac{1}{2x}\right)^{8-r} = -x^1$
 $2r + -1(8-r) = 1$
 $2r - 8 + r = 1$
 $3r = 9$
 $r = 3 \checkmark$

Coefficient $= {}^8C_3 \left(\frac{1}{2}\right)^{8-3}$
 $= {}^8C_3 \left(\frac{1}{2}\right)^5$
 $= 1.25 \checkmark$

c) i) $(0.95)^{20} \doteq 0.36 \quad .35^8 \checkmark$

ii) ${}^{20}C_4 (0.05)^4 (0.95)^{16} \doteq 0.013 \checkmark$

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Q5(a)(i)

$$\begin{aligned}\int_{\pi/4}^{\pi/2} \cos^2 x \, dx &= \frac{1}{2} \int_{\pi/4}^{\pi/2} (\cos 2x + 1) \, dx \\ &= \frac{1}{2} \left[\frac{1}{2} \sin 2x + x \right]_{\pi/4}^{\pi/2} \\ &= \frac{1}{2} \left[\left(\frac{1}{2} \sin \pi + \frac{\pi}{2} \right) - \left(\frac{1}{2} \sin \frac{\pi}{2} + \frac{\pi}{4} \right) \right] \\ &= \frac{1}{2} \left(\frac{\pi}{2} - \frac{1}{2} - \frac{\pi}{4} \right) \\ &= \frac{\pi}{8} - \frac{1}{4}\end{aligned}$$

(2) generally well done.

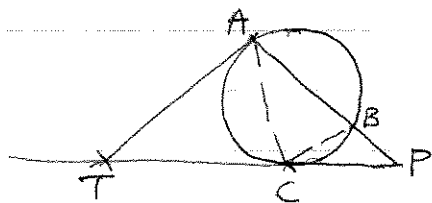
(ii) Vol = $\pi \int_{\pi/4}^{\pi/2} (1 - \cos x)^2 \, dx$

$$\begin{aligned}&= \pi \int_{\pi/4}^{\pi/2} 1 - 2\cos x + \cos^2 x \, dx \\ &= \pi \left[x - 2\sin x + \frac{x}{2} - \frac{1}{4} \right]_{\pi/4}^{\pi/2} \\ &= \pi \left(\left(\frac{\pi}{2} - 2 \right) - \left(\frac{\pi}{4} - \sqrt{2} \right) + \frac{\pi}{8} - \frac{1}{4} \right) \\ &= \pi \left(\frac{3\pi}{8} - 2\frac{1}{4} + \sqrt{2} \right)\end{aligned}$$

some did not use part (i). Many messed up substitution.

(2)

Q5(b)



(i) $\hat{BAC} = \hat{BCP}$ (alt segment theorem) ✓
 $\hat{BAT} = 90^\circ$ (angle between tangent and diameter) ✓
 $\therefore \hat{CAT} = \hat{BAT} - \hat{BAC}$
 $= 90^\circ - \hat{BCP}$

many answers lacked sequence or had poor explanations.

(2)

(ii) Since $AT = CT$ (tangents from external point to a circle are equal length) ✓

$\therefore \triangle ATC$ is isosceles.

hence $\hat{ACT} = \hat{CAT} = 90^\circ - \hat{BCP}$ (base angles of $\triangle$)

$\therefore \hat{ATC} = 180^\circ - 2(90^\circ - \hat{BCP})$ (sum of angles in $\triangle$)

$= 2\hat{BCP}$ //

(2)

5(c)(i) $\frac{dT}{dt} = -kAe^{-kt}$

$= -k(T - 20)$ (since $Ae^{-kt} = T - 20$) ✓ (1)

(ii) $t = 0$, $T = 90$ h/a $A = 70$ ✓

$T = 20 + 70e^{-kt}$

When $T = 70$, $t = 10$

$70 = 20 + 70e^{-10k} \Rightarrow k = -\frac{1}{10} \ln \frac{5}{7}$ ✓

When $T = 65$, $T = 20 + 70e^{-\frac{1}{10} \ln \frac{5}{7} \times 15}$

$= 62^\circ$ (nearest degree) ✓

(3)

well done

Question 6

$$\begin{aligned} \text{a) i) } f'(x) &= \frac{e^x(1+e^x) - e^x \cdot e^x}{(1+e^x)^2} \\ &= \frac{e^x + e^{2x} - e^{2x}}{(1+e^x)^2} \\ &= \frac{e^x}{(1+e^x)^2} \end{aligned}$$

> 0 for all values of x

$\therefore f(x)$ always has a positive gradient

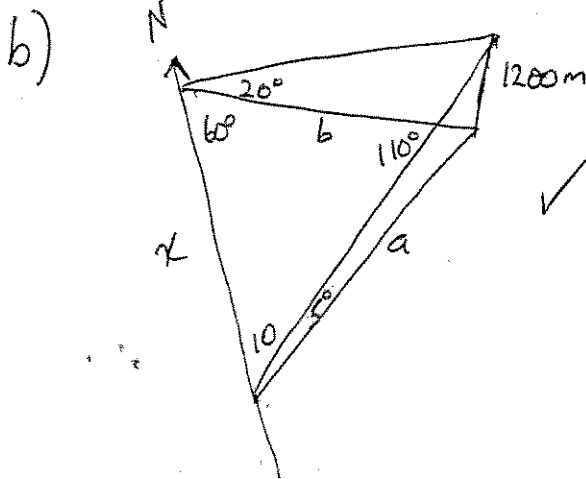
$\therefore f(x)$ is increasing for all values of x

$$\begin{aligned} \text{iii) } x &= \frac{e^y}{1+e^y} \\ (1+e^y)x &= e^y \\ x + xe^y &= e^y \\ xe^y - e^y &= -x \end{aligned}$$

ii) Since $f(x)$ is always increasing for each y value there can only be one x value.

$$\begin{aligned} e^y(x-1) &= -x \\ e^y &= \frac{-x}{x-1} \end{aligned}$$

$$y = \ln\left(\frac{-x}{x-1}\right)$$



$$\begin{aligned} \tan 5^\circ &= \frac{1200}{a} & \tan 20^\circ &= \frac{1200}{b} \\ a &= \frac{1200}{\tan 5^\circ} & b &= \frac{1200}{\tan 20^\circ} \end{aligned}$$

$$x^2 = \left(\frac{1200}{\tan 5^\circ}\right)^2 + \left(\frac{1200}{\tan 20^\circ}\right)^2 - 2\left(\frac{1200}{\tan 5^\circ}\right)\left(\frac{1200}{\tan 20^\circ}\right)\cos 110^\circ$$

$$x = 15163.56578 \text{ m}$$

$$\text{Speed} = \frac{15.163}{1\frac{1}{2}} \div 10.1 \text{ km/h}$$

c) $\sin x + \sqrt{3} \cos x$

$$2\left(\frac{1}{2} \sin x + \frac{\sqrt{3}}{2} \cos x\right)$$

$$2(\cos A \sin x + \sin A \cos x)$$

$$2\left(\sin\left(x + \frac{\pi}{3}\right)\right)$$

$$\sin\left(x + \frac{\pi}{3}\right) \leq \frac{1}{2}$$

$$x + \frac{\pi}{3} \leq \frac{\pi}{6} \text{ or } \frac{5\pi}{6}$$

$$x = \frac{\pi}{6} \text{ or } \frac{\pi}{2}$$



7.(a)(i) $y=0$ when rocket hits ground

$$\therefore -\frac{1}{2}gt^2 + Vt \sin \theta = 0$$

$$\therefore t(-\frac{1}{2}gt + V \sin \theta) = 0$$

$$\therefore \text{either } t=0 \text{ or } \frac{1}{2}gt = V \sin \theta$$
$$t = \frac{2V \sin \theta}{g}$$

(ii) when $t = \frac{2V \sin \theta}{g}$:

$$x = \frac{2V^2 \sin \theta \cos \theta}{g}$$

(1) ✓

(iii) for the boat: $x = \frac{Vt}{2}$

$$\therefore x = \frac{2V^2 \sin \theta}{2g}$$

(2) ✓

Equating (1) and (2) :

$$\frac{2V^2 \sin \theta \cos \theta}{g} = \frac{2V^2 \sin \theta}{2g}$$

$$\therefore 2 \cos \theta = 1$$

$$\therefore \cos \theta = \frac{1}{2}$$

$$\therefore \theta = \frac{\pi}{3} \quad (\text{must be in first quadrant})$$

(b) To prove: $9^n - 3$ is divisible by 6 for all positive integers n

$$\text{For } n=1 : 9^1 - 3 = 6 \\ \therefore \text{true for } n=1$$

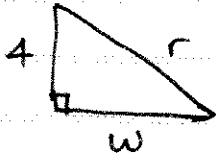
$$\text{Assume true for } n=k \\ \therefore 9^k - 3 = 6M \text{ where } M \text{ is an integer} \quad \checkmark$$

Prove true for $n=k+1$

$$\begin{aligned} 9^{k+1} - 3 &= 9 \cdot 9^k - 3 \\ &= 9(6M + 3) - 3 \quad \checkmark \\ &= 54M + 27 - 3 \\ &= 54M + 24 \\ &= 6(9M + 4) \quad \checkmark \\ \therefore \text{true for } n=k+1 \end{aligned}$$

Since it is true for $n=1$, and when we assume that it is true for $n=k$ then it follows that it is also true for $n=k+1$, then it must be true for all integers greater than or equal to 1.

(c)



$$r^2 = 16 + w^2$$

$$r = \sqrt{16 + w^2}$$

✓

$$\therefore \frac{dr}{dw} = \frac{1}{2} \times 2w \times (16 + w^2)^{-1/2}$$

$$= \frac{w}{\sqrt{16 + w^2}}$$

$$\frac{dr}{dt} = 2$$

$$\begin{aligned} \therefore \frac{dw}{dt} &= \frac{dw}{dr} \times \frac{dr}{dt} \\ &= \frac{\sqrt{16 + w^2}}{w} \times 2 \end{aligned}$$

✓

when $r = 10$, ~~100~~ $100 = 16 + w^2$

$$\therefore w = \sqrt{84}$$

✓

$$\therefore \frac{dw}{dt} = \frac{\sqrt{16 + 84}}{\sqrt{84}} \times 2$$

$$= \frac{20}{\sqrt{84}}$$

✓

2.18